Stellar Cyber 6.6.0s is now live! Discover all the updates in the release notes!

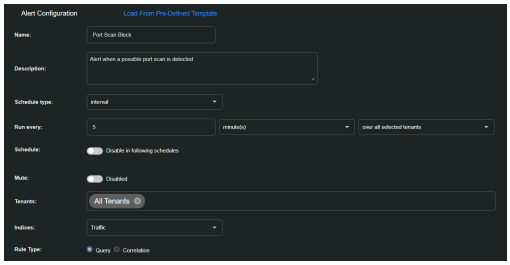
ATH Example: Blocking a Port Scanning Source IP

Create an Automated Threat Hunting (ATH) playbook that notifies you when a source IP address is detected pinging a specified destination IP address on more than 20 destination ports. This is known as port scanning and is a common technique used to identify open ports on a target host. While port scans are often used during legitimate security assessments, they can also indicate hostile reconnaissance. Any alert triggered by this playbook requires further investigation, such as checking whether the source IP address belongs to a known vulnerability scanner or an unrecognized external system.

This example also includes instructions on how to use nested calculations in an ATH playbook.

Configure the Alert

Create an alert that runs a query every 5 minutes on the **Traffic** index.



1. Navigate to **Automation | Automation** and select the **Create** button to add a playbook.
2. Enter the following in the **Alert Configuration** section:

   Name: **Port Scan Block**

   Description: **Alert when a possible port scan is detected**

   Schedule type: **interval**

   Run every: **5 minute(s) over all selected tenants**

   Schedule: Toggle off (default setting)

   Mute: Toggle off (default setting)

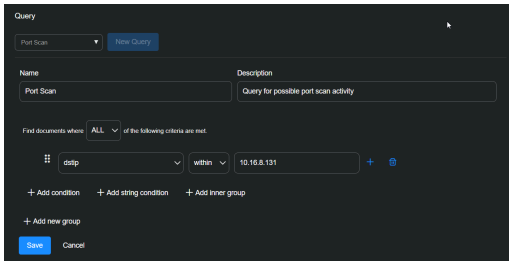
Tenants: **All Tenants**

Indices: **Traffic**

Rule Type: **Query**

Build a Query

Create a query that looks for new records with a **dstip** of **10.16.8.131**. This is the IP address that appears to be the target of port scans.



1. Select **New Query**.
2. Enter the following:

   Query Name: **Port Scan**

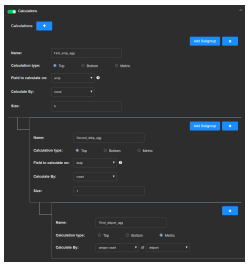
   Description: **Query for possible port scan activity**

   Find documents where **ALL** of the following criteria are met.

   Select **+ Add condition** and enter **dstip**, **within**, **10.16.8.131**.
3. Select **Save** to save your query.

Add Calculations to the Query

In the **Calculations** section, build a series of nested calculations to match the top five source IP addresses that are probing 10.16.8.131 on more than 20 destination ports. This behavior is characteristic of intentional port scanning, where a source attempts to discover open or responsive services on a target host by sweeping across multiple ports.



1. Toggle **Calculations** on.
2. Select **+** to add a calculation.
3. Enter the following to match the top five source IP addresses probing 10.16.8.131:

   Name: **First_srcip_agg**

   Calculation type: **Top**

   Field to calculate on: **srcip**

   Calculation By: **count**

Size: **5**

4. Select **Add Subgroup** and add a nested calculation for the top destination IP address:

   Name: **Second_dstip_agg**

   Calculation type: **Top**

   Field to calculate on: **dstip**

   Calculate By: **count**

   Size: **1**

   > Because this query is for just one destination IP address (**10.16.8.131**), the size is **1**. If your query is for multiple destination IP addresses, you can set this to a larger size.

5. Select **Add Subgroup** to add a third nested calculation that matches source IP addresses probing more than 20 destination ports.

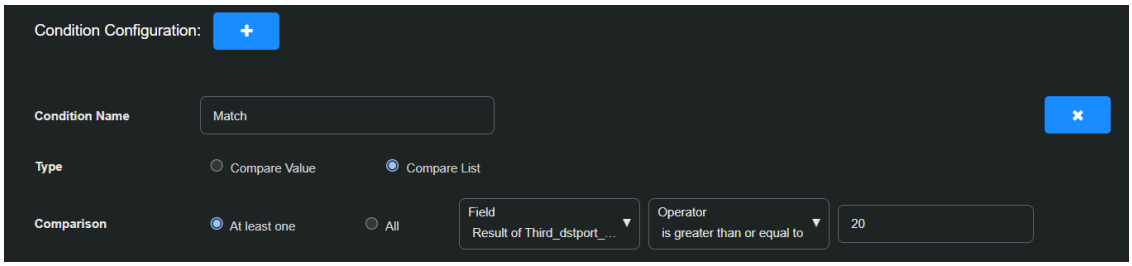
   Name: **Third_dstport_agg**

   Calculation type: **Metric**

   Calculate By: **unique count** of **dstport**

Configure a Condition

To look for hosts probing 10.16.8.131 on more than 20 ports, add a condition that triggers if the count of the third calculation (**Result of Third_dstport_agg**) is greater than or equal to **20**.



Condition Name: **Match**

Type: **Compare List**

Comparison: **At least one**, **Result of Third_dstport_agg**, **is greater than or equal to**, **20**

Configure Actions

Finally, you must configure a Firewall/Security Switch action to block the IP address when the **Match** condition you configured in the previous section is detected. Refer to Configuring Firewall/Security Switch Actions for details on configuring this type of action.

When you have finished, select **Submit** to save and immediately run the playbook.

The new playbook appears in the Automated Threat Hunting Playbooks list.



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